Doing More with Less: An Adaptive, Label-Efficient Approach to Fraud Detection from Day One

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Agenda

- Why Smarter Detection Can't Wait
- Background
- Our Solution: A Smarter, Adaptive Learning Pipeline
  - X-ITERADE
    - Explainability via X-ITERADE
  - ALISA
- Results
- Conclusion
- Q&A

Why This Matters: The Rising Cost of Fraud

- Global fraud losses are exploding:
  - $44.3B in eCommerce fraud in 2024
  - Predicted to reach $48B by 2025 (16% YoY increase)
  - In the U.S. alone, fraud losses exceeded $10B in 2023 (14% increase from 2022)
  - Canadians lost over $638M to fraud in 2024, but only 5–10% of frauds are reported

- Fraud fuels terrorism, trafficking, and organized crime.

- AI is powering smarter fraud; detection must keep pace.

- Fraud is no longer just a financial issue; it's a global security problem

Why Smarter Fraud Detection Can't Wait

- Fraud is severely underreported; the true scale is unknown
- Most detection systems still rely on outdated, rigid rules
- Traditional methods struggle with new, fast-evolving fraud tactics
- Industry response is shifting:
  - 75% of eCommerce firms will boost fraud prevention budgets in 2025
  - Many increasing investment by 20% or more

Challenges in Fraud Detection

- Extreme class imbalance: Fraud is <1% of all transactions
- Label scarcity: Experts are expensive and slow, and data is sensitive
  - Labeled data is scarce, or entirely missing
  - Often only one class is labeled (usually non-fraud)
- High cost of mistakes:
  - False negatives = undetected fraud
  - False positives = frustrated customers
- Explainability is critical: Black-box models don't build trust

Background

- AL vs. SSL
- Both help reduce label cost, but…
  - Fail in extremely imbalanced or one-class cases
  - Require tuning, large budgets, and clean starting points

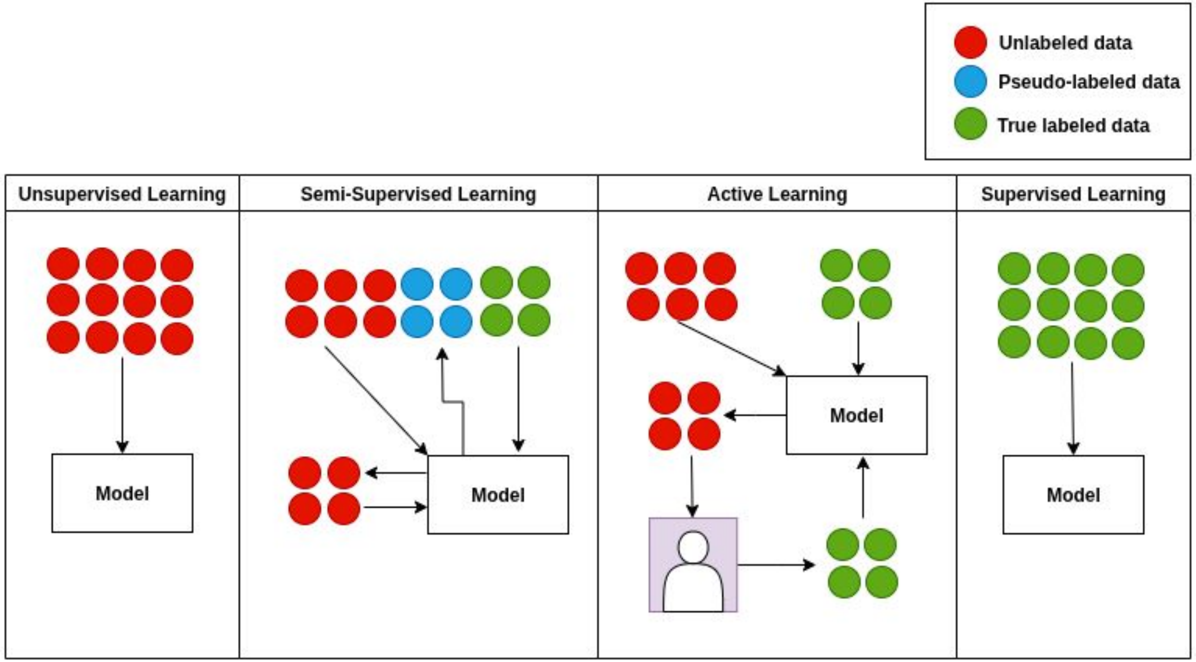

Figure 1: How Learning Paradigms Differ by Label Availability.

Our Solution: A Smarter, Adaptive Learning Pipeline

- Stage 1: X-ITERADE
  - Fully unsupervised anomaly detection
  - Selects a high-quality subset of suspicious and safe cases
  - Includes explainability via SHAP + LLMs

- Stage 2: ALISA
  - Builds on X-ITERADE's output
  - Combines AL + SSL + Data Augmentation
  - Adapts based on model feedback and data properties

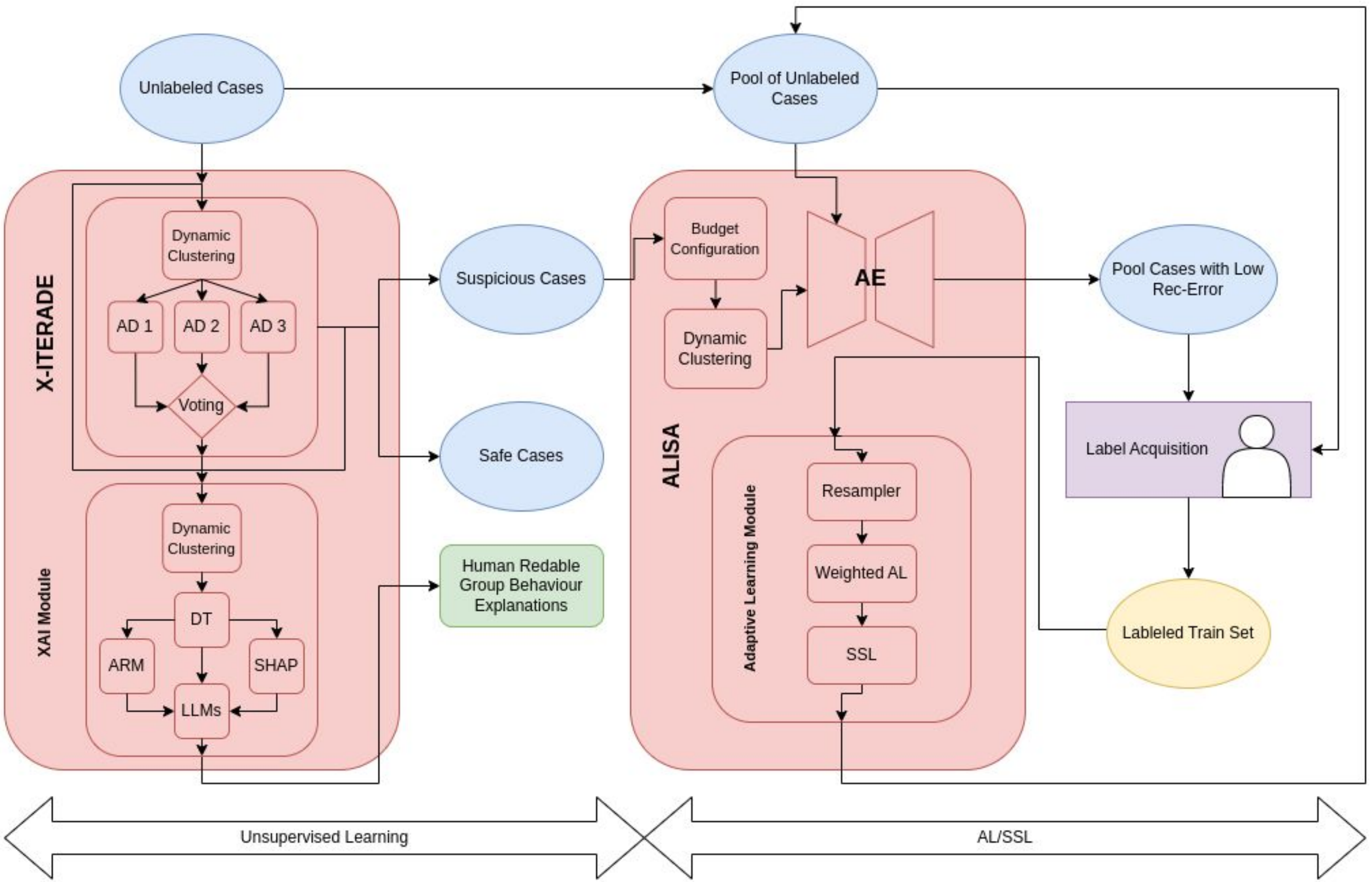

Figure 2: Overview of the proposed two-stage framework.

Step 1 - X-ITERADE

- Explainable Iterative Anomaly Detection Ensemble with dynamic clustering
- Goal:
  - Iteratively separates anomalous cases from normal ones and improves the imbalance ratio in a completely unsupervised manner
- Benefits:
  - Completely unsupervised
  - Helps experts prioritize labeling
  - Flexible solution for various labeling budgets
  - Outperforms existing baselines

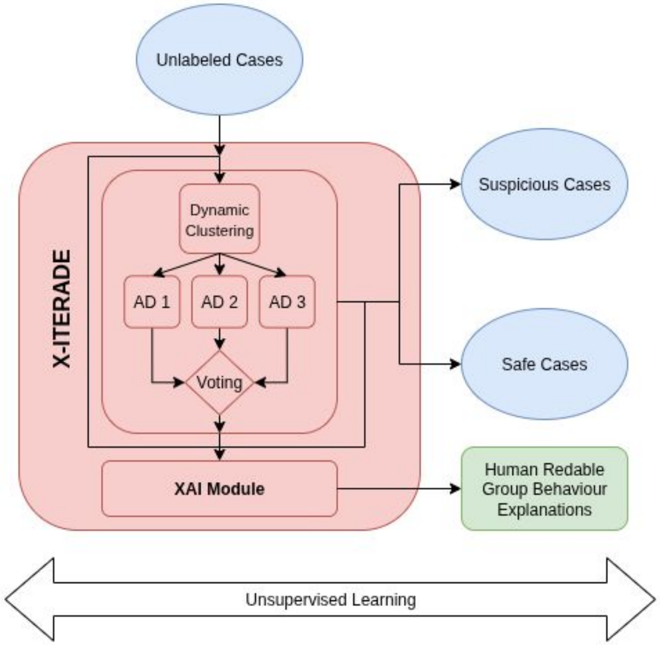

Figure 3: Overview of X-ITERADE.

Explainability in X-ITERADE

- Objective: Translate anomalies into clear, group-based behavioral profiles to support trust and action
  - For analysts, clients, and researchers

- LLM Test Example:
  - Key SHAP Values: Zip (high impact), Has Chip (low impact)
  - Association Rules:Has Chip = YES ⇒ Card Brand = Mastercard
  - Distribution Info: 3057 non-fraud, 29 fraud (highly suspicious)
  - Additional Context: Has Chip = YES, Zip > 79970.50, Total Transactions ≤ 169.00

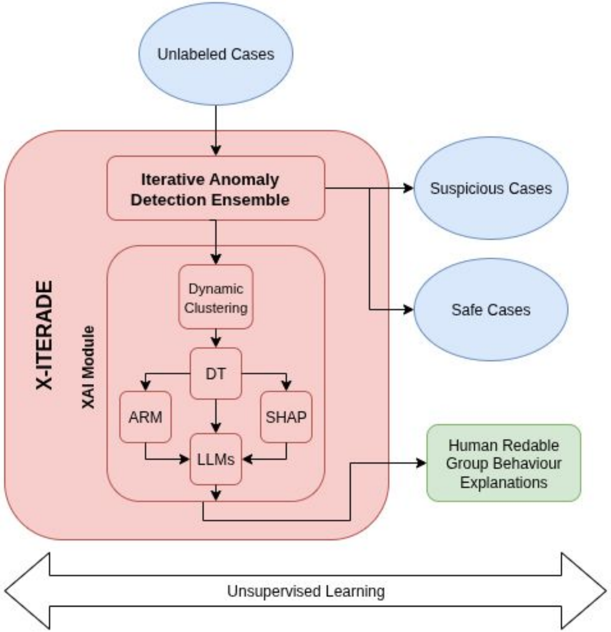

Figure 4: Overview of XAI Module.

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Step 2 - ALISA

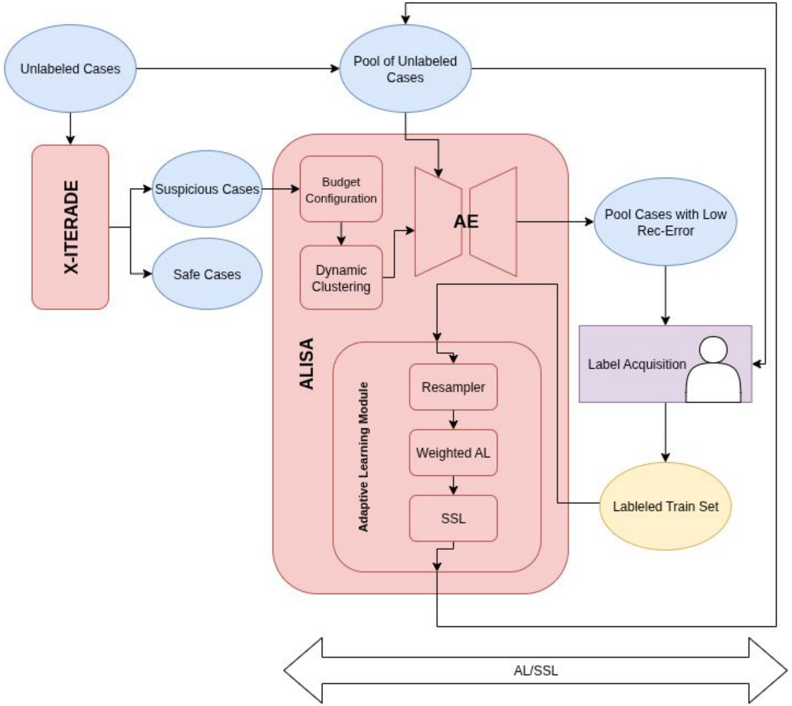

Figure 5: Overview of ALISA.

- Goal:
  - Expand labeled data intelligently while reducing labeling cost
- Works from minimal or one-class labels
- Combines 5 key components:
  - Budget-aware configuration
  - Autoencoder-based selection per cluster
  - Data augmentation
  - AL
  - SSL
- Iteratively switches between AL and SSL based on performance
- Supports sequential or parallel execution
- Learns what works best and adjusts weights dynamically

X-ITERADE Key Results

- Impact:
  - Fraud-to-anomalies ratio increased significantly.
  - Positives identified **3×** to **15×** more than baselines.

Algorithm	Performance		Improv.	Average Anomalies	
	Precision	F1 Score		TP	Anom. Det
HDBSCAN	0.0011	0.0022	0.786	4	3647
X-ITERADE - SV 2.1 - MR	0.00972	0.01925	**6.943**	12	1234
X-ITERADE - HV 1.2 -MR	0.00328	0.00654	2.343	6.5	1983
X-ITERADE - SV 2.3	0.00622	0.01236	4.443	18.33	2947.66

Table 1: ITERADE average results on CCT-Original dataset form 600<budget<4000 compared to the baselines.

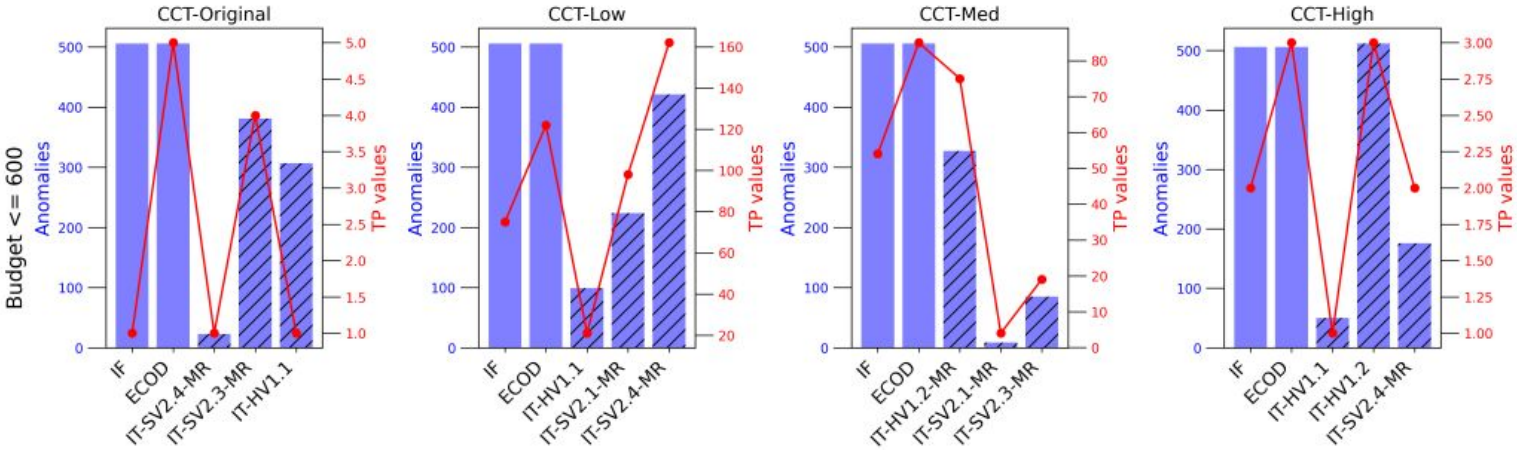

Figure 6: Comparison baselines with best ITERADE results on CCT dataset for the budget <= 600.

XAI Module Key Results

- LLM Output:
  - This group is characterized by users with **chipped cards**, primarily using **Mastercard**, and residing in zip codes greater than 79970.50. Despite the low impact of the chip feature, the high SHAP value for zip code and the presence of 29 fraud cases out of 3086 total transactions highlight a **significant fraud risk**. The combination of these factors suggests that **further investigation is warranted** to understand potential vulnerabilities associated with the geographic location and transaction volume.

- **Mistral 7B Instruct v0.3 ≈ GPT-4o** in performance, outperforming LLaMA 3.1 Instruct.

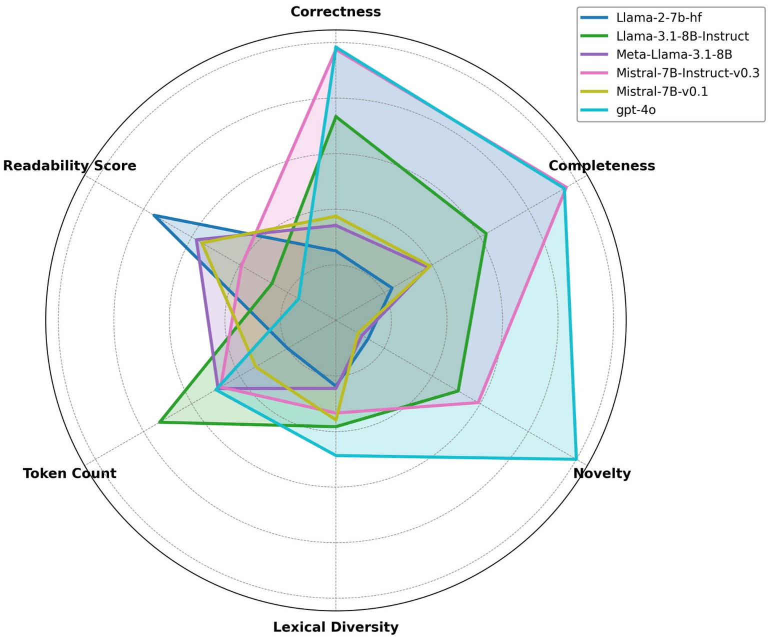

Figure 7: Comparison of LLM average performance across 20 test examples and 3 human annotators on the CCT Dataset.

ALISA Key Results

- Performance Gains:
  - +5–22% F1 score
  - +60–80% precision over the best baseline

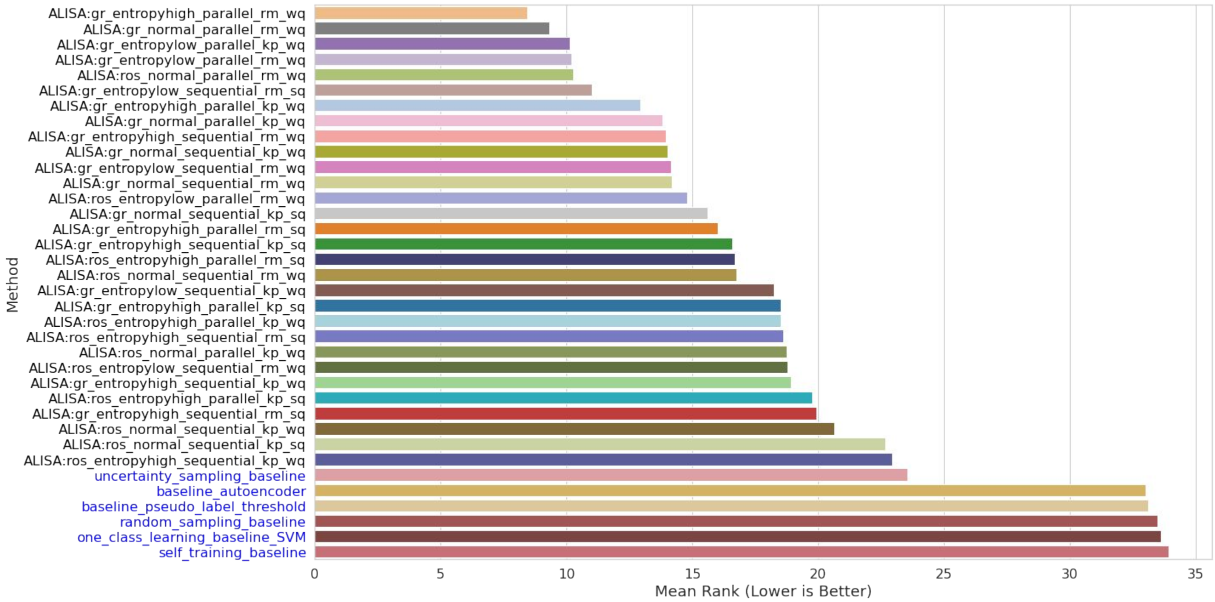

Figure 8: Average Rank of ALISA Methods and the Baselines on CCT Datasets (Lower Values Indicate Better Performance)

Summary & Key Takeaways

Objective: Tackle fraud detection with extremely limited or no labeled data.

Approach:

- Start fully unsupervised → gradually add AL & SSL.
- Progression reflects real-world data availability.

Key Contributions:

- **X-ITERADE** – iterative unsupervised anomaly detection with budget control & explainability.
- **ALISA** – modular AL+SSL+augmentation pipeline, excelling in label-scarce scenarios.

Impact: Up to **22% F1** improvement over best baselines; supports transparency via explainability module.

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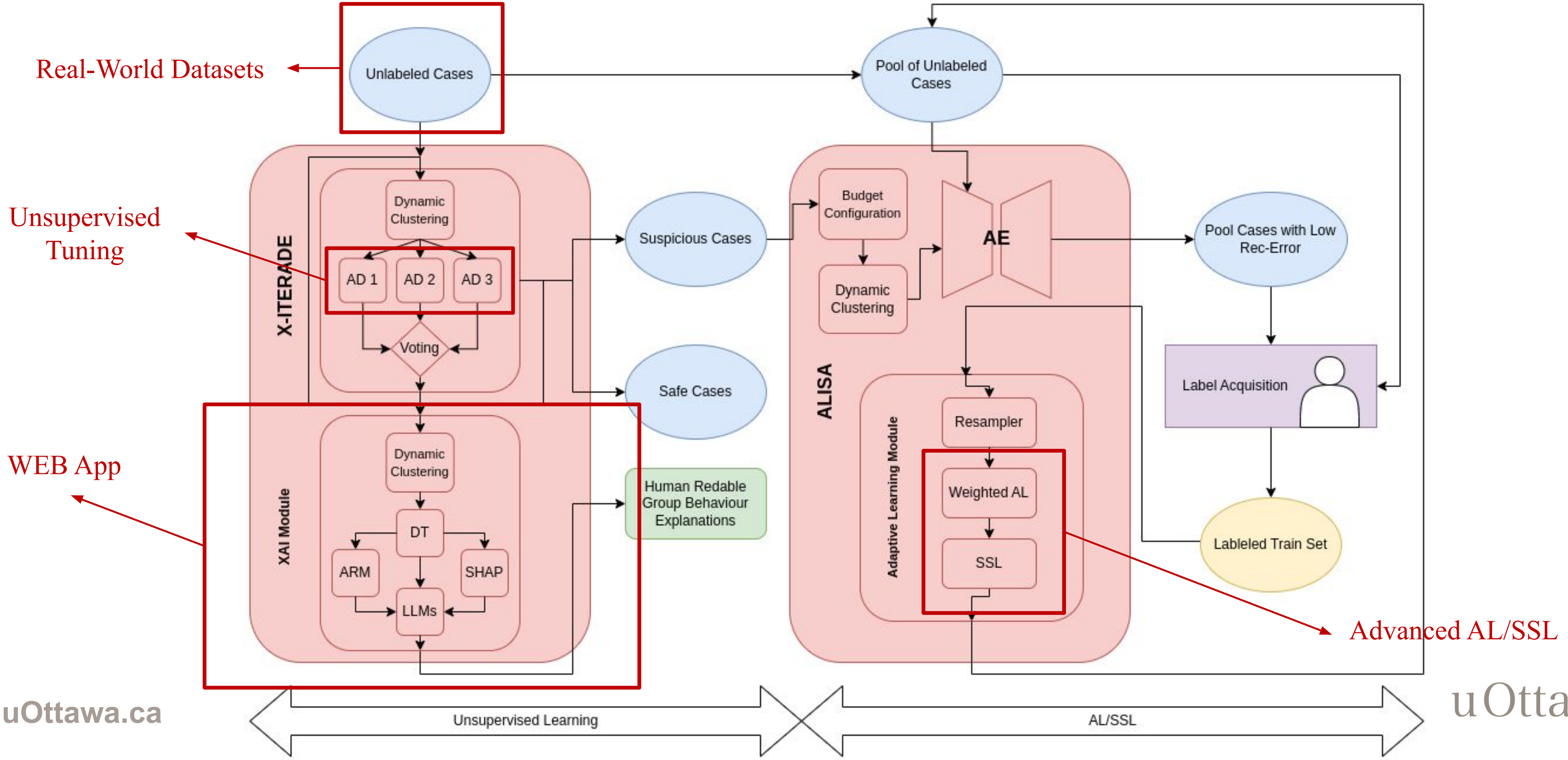
Now & Next



Conclusion

- **Proposed:** Fully unsupervised start with **X-ITERADE**, then progressively add **AL & SSL** with **ALISA** to adapt as labels appear.

- **Strengths:** Scalable, explainable, and cost-efficient; balances accuracy, efficiency, and transparency under real-world constraints.

- **Impact:** Bridges research and deployment, evolving with data to cut labeling costs and support informed decisions.

Thank you!

Questions?

Paper

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